NANODE · FIREWORKS HACKATHON · NOVEMBER 2, 2026

Glimmer: base to better

One LoRA training run on Nanode build traces, then the same model before and after, building the same mesh.

597
training examples

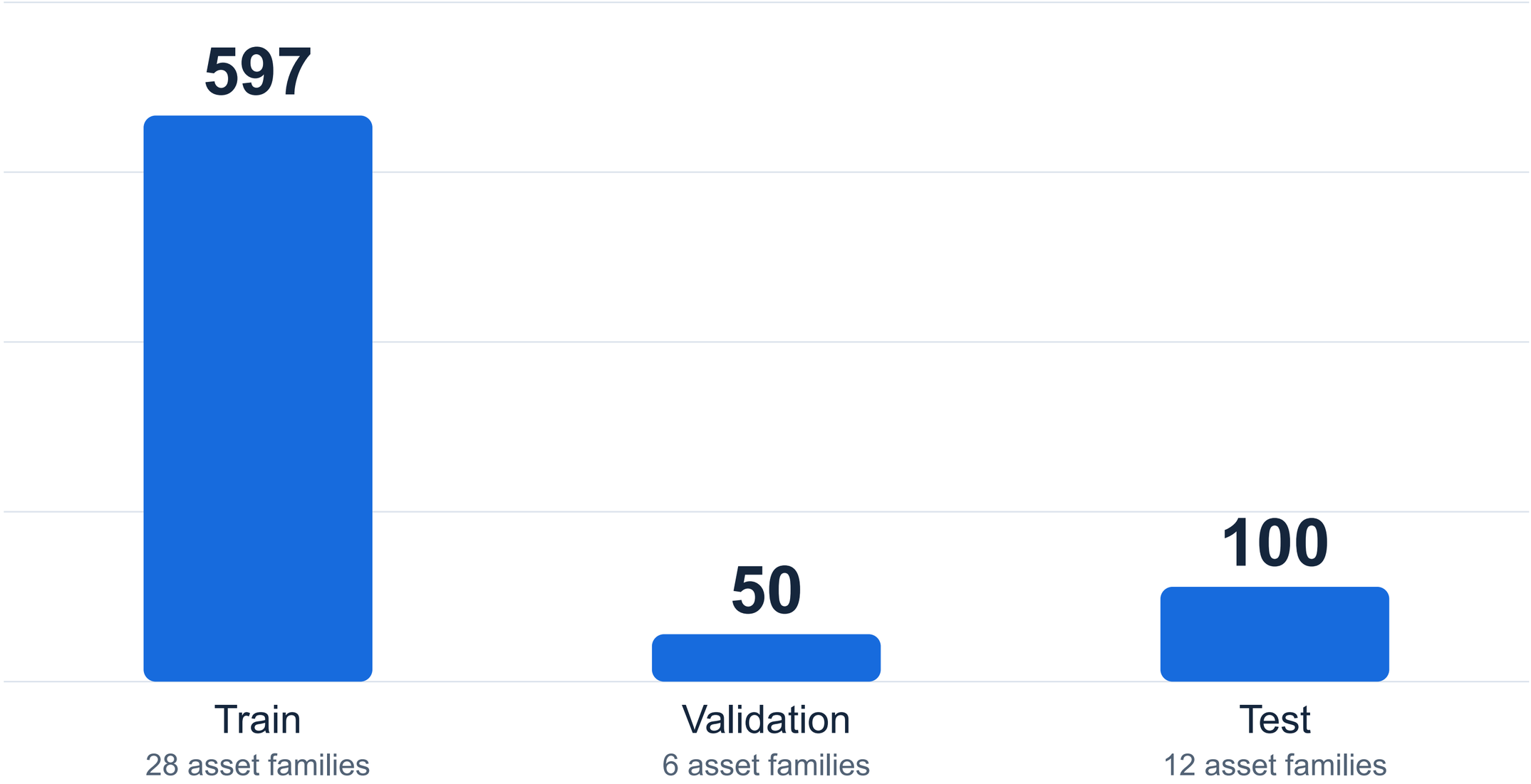
14m 58s
training wall time

100 pairs
held-out base vs LoRA

4 meshes
base / LoRA × iteration 1 / 5

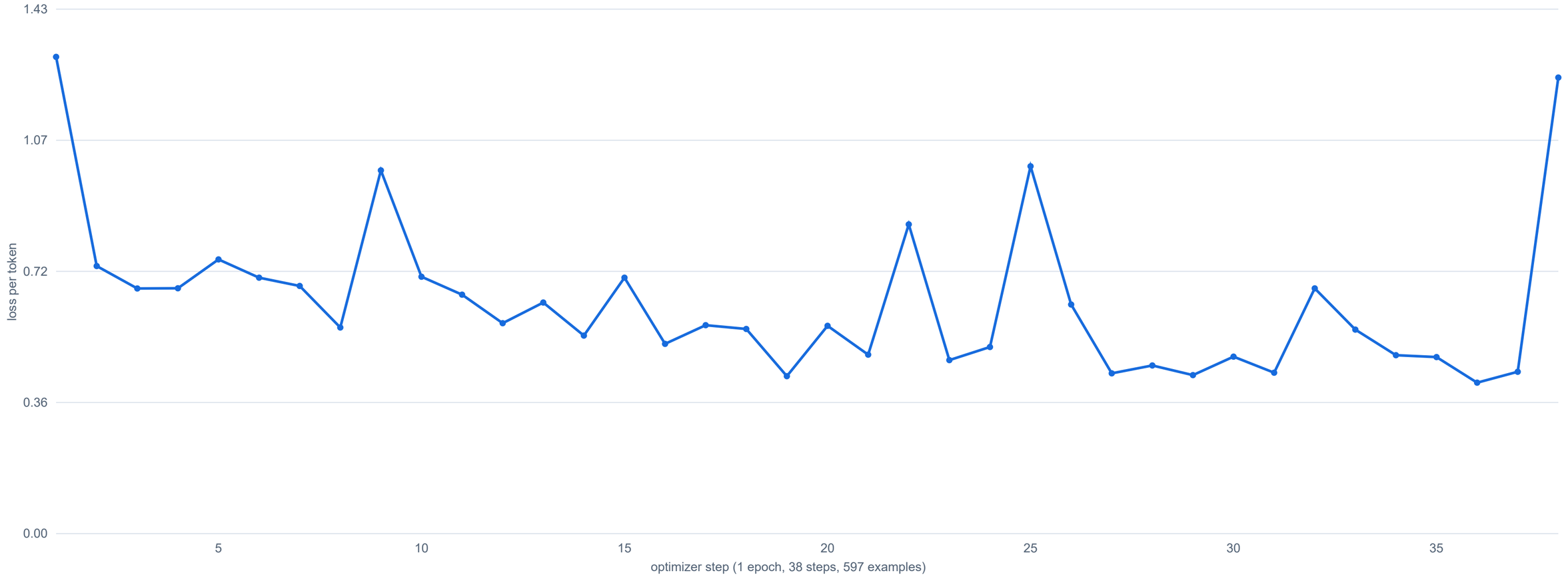
Sources: data/preflight_report.json · progress/training-job.receipt.json · data/test-summary.json · hero-runtime-lock.json

166 Nanode build jobs → 747 workflow actions, split by asset family



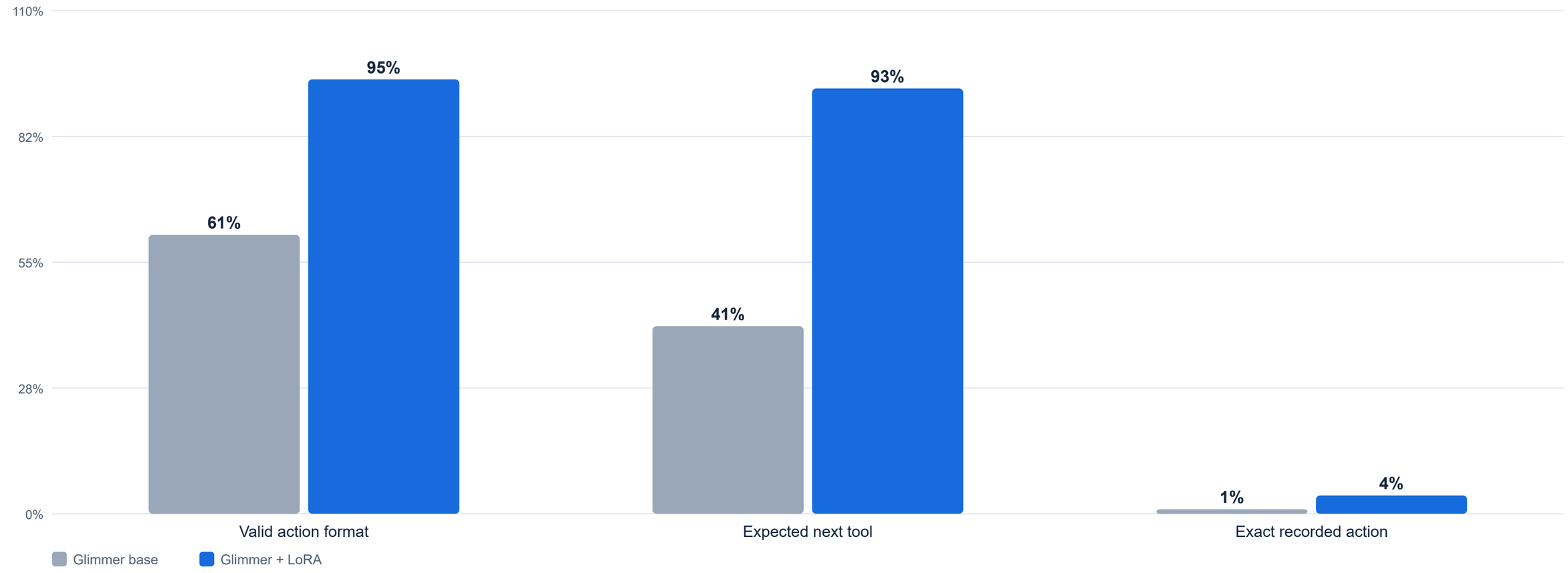
Sources: aws_probe/trace_catalog_20261102.json · data/preflight_report.json

Training: Glimmer 30B, LoRA rank 8, 1 epoch, 38 steps, 14m 58s



Sources: progress/training-state.snapshot.json · progress/training-metrics.snapshot.jsonl · progress/training-job.receipt.json

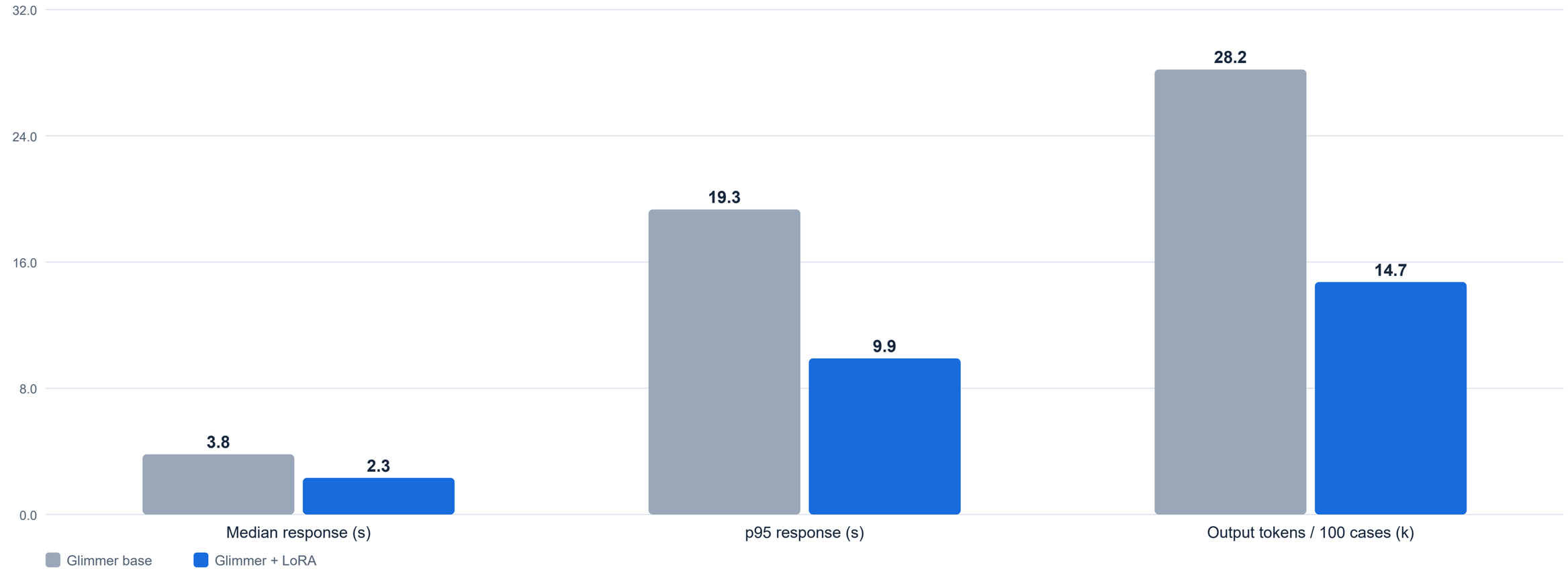
100 held-out pairs: the LoRA follows the workflow far more often



Same prompts, temperature 0. Action-format scores, not mesh quality. Exact-action gain is uncertain (family interval crosses zero).

Sources: data/test-summary.json · data/paired-results-analysis.json

39% lower median response time, about half the output



Sources: data/test-summary.json

Budget: $35.65 reserved of $50, $0.00 billed; meshes so far: 0 of 4



v1 base
controller_or_adapter_error

v1 LoRA
step_limit

v2 base
iteration_target_not_met

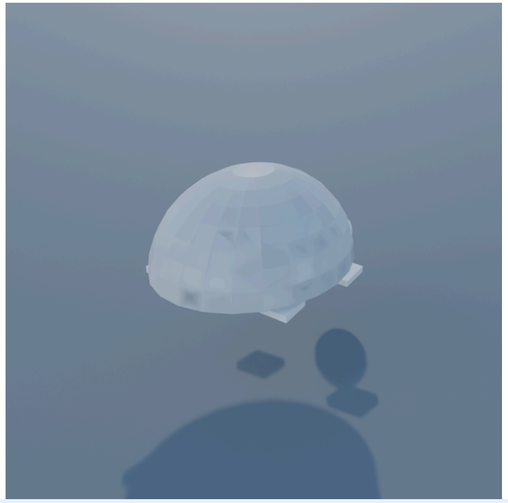
v2 LoRA
context_limit

Reservations are 2× list rate and never refunded (upper bound). GLM 5.3 Flash in parallel: no-go, ≈ $94.38 training. Build-only harness v4 is producing the meshes on the next slides.

Sources: budget_policy.json · hero-results/tuned-v2/budget-summary.json · data/billing-snapshot.json · hero-results/*/hero-*.case_receipt.json · GLM_PARALLEL_FEASIBILITY.md

Mesh comparison: base vs LoRA, iterations 1, 2 and 5

Armored tortoise courier robot (0.60 × 0.90 × 0.65 m), same brief, same sandbox, same camera. Front 3/4 view.

	iteration 1	iteration 2	iteration 5
Glimmer base	 Glimmer base · iteration 1 · Front 3/4	 Glimmer base · iteration 2 · Front 3/4	 Glimmer base · iteration 5 (latest available: iteration 3) · Front 3/4
Glimmer + LoRA	RENDER PENDING glimmer_meshes_20261102\tuned_iter1\renders\front34.png Glimmer + LoRA · iteration 1 · Front 3/4	RENDER PENDING glimmer_meshes_20261102\tuned_iter2\renders\front34.png Glimmer + LoRA · iteration 2 · Front 3/4	RENDER PENDING glimmer_meshes_20261102\tuned_iter5\renders\front34.png Glimmer + LoRA · iteration 5 · Front 3/4

Sources: hero-runtime-lock.json · W:/Temp/opencode/glimmer_meshes_20261102/{base,tuned}_iter{1,2,5}/renders/front34.png

Base Glimmer, iteration 1: invented a Blender option

build.py, line 108

```
bmesh.ops.create_cone(bm, cap_verts=8, cap_tris=8,
    radius1=0.03, radius2=0.01, depth=0.06)
```

Blender 5.2.2

```
TypeError: create_cone: keyword "cap_verts" is invalid for this
operator
```

The real keyword is **segments**. The script stopped there with 7 objects built: Head, Plate_0 to Plate_4, Shell_Dome (Shell_Dome flagged as an invalid mesh). Legs, tail and gear were never built.

Base, iteration 1: stopped with an error after exporting 7 objects

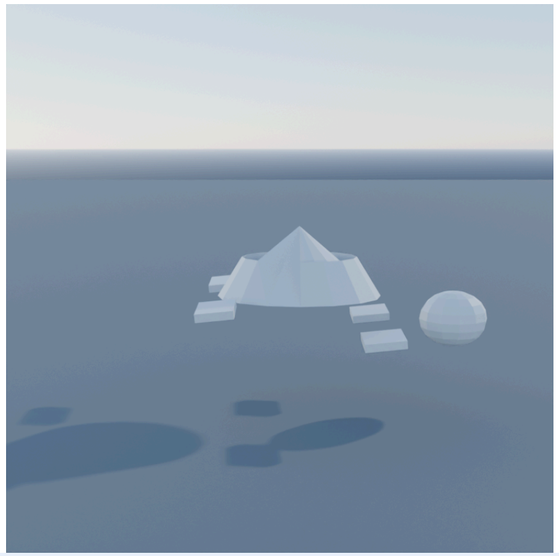
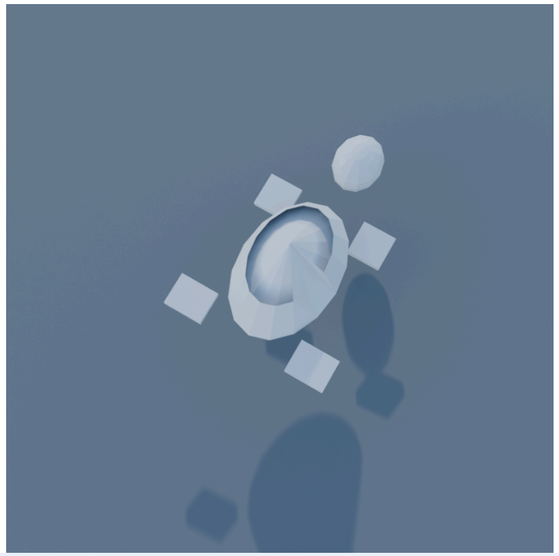
LoRA result pending



Glimmer base · iteration 1 · Front 3/4

Sources: W:/Temp/opencode/glimmer_meshes_20261102/base_iter1/{build.py,checks.json,note.txt,renders/front34.png} · W:/Temp/opencode/glimmer_meshes_20261102/tuned_iter1/checks.json

Iteration 1: base vs LoRA

Glimmer base			
	Glimmer base · iteration 1 · Front 3/4	Glimmer base · iteration 1 · Side	Glimmer base · iteration 1 · Top
Glimmer + LoRA	RENDER PENDING glimmer_meshes_20261102\tuned_iter1\renders\front34.png	RENDER PENDING glimmer_meshes_20261102\tuned_iter1\renders\side.png	RENDER PENDING glimmer_meshes_20261102\tuned_iter1\renders\top.png
	Glimmer + LoRA · iteration 1 · Front 3/4	Glimmer + LoRA · iteration 1 · Side	Glimmer + LoRA · iteration 1 · Top

Sources: W:/Temp/opencode/glimmer_meshes_20261102/{base,tuned}_iter1/renders/{front34,side,top}.png

Iteration 5: base vs LoRA

Glimmer base



Glimmer base · iteration 5 (latest available: iteration 3) · Front 3/4



Glimmer base · iteration 5 (latest available: iteration 3) · Side



Glimmer base · iteration 5 (latest available: iteration 3) · Top

Glimmer + LoRA

RENDER PENDING

`glimmer_meshes_20261102\tuned_iter5\renders\front34.png`

Glimmer + LoRA · iteration 5 · Front 3/4

RENDER PENDING

`glimmer_meshes_20261102\tuned_iter5\renders\side.png`

Glimmer + LoRA · iteration 5 · Side

RENDER PENDING

`glimmer_meshes_20261102\tuned_iter5\renders\top.png`

Glimmer + LoRA · iteration 5 · Top

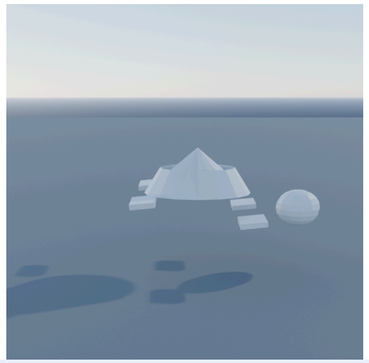
Sources: W:/Temp/opencode/glimmer_meshes_20261102/{base,tuned}_iter5/renders/{front34,side,top}.png

Glimmer base: iterations 1, 2 and 5

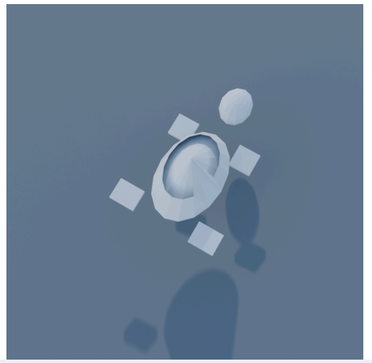
iteration 1



Glimmer base · iteration 1 · Front 3/4



Glimmer base · iteration 1 · Side

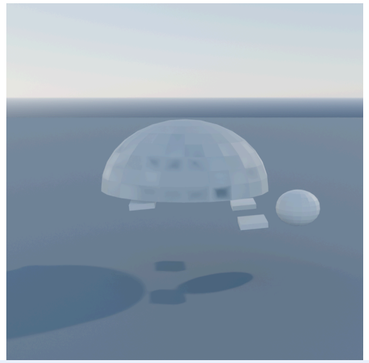


Glimmer base · iteration 1 · Top

iteration 2



Glimmer base · iteration 2 · Front 3/4



Glimmer base · iteration 2 · Side

NO RENDER

top.png missing in base_iter2

Glimmer base · iteration 2 · Top

iteration 5



Glimmer base · iteration 5 (latest available: iteration 3) · Front 3/4



Glimmer base · iteration 5 (latest available: iteration 3) · Side



Glimmer base · iteration 5 (latest available: iteration 3) · Top

Sources: W:/Temp/opencode/glimmer_meshes_20261102/base_iter{1,2,5}/renders/{front34,side,top}.png

Glimmer + LoRA: iteration 1 vs 5

iteration 1	RENDER PENDING glimmer_meshes_20261102\tuned_iter1\renders\front34.png	RENDER PENDING glimmer_meshes_20261102\tuned_iter1\renders\side.png	RENDER PENDING glimmer_meshes_20261102\tuned_iter1\renders\top.png
	Glimmer + LoRA · iteration 1 · Front 3/4	Glimmer + LoRA · iteration 1 · Side	Glimmer + LoRA · iteration 1 · Top
iteration 5	RENDER PENDING glimmer_meshes_20261102\tuned_iter5\renders\front34.png	RENDER PENDING glimmer_meshes_20261102\tuned_iter5\renders\side.png	RENDER PENDING glimmer_meshes_20261102\tuned_iter5\renders\top.png
	Glimmer + LoRA · iteration 5 · Front 3/4	Glimmer + LoRA · iteration 5 · Side	Glimmer + LoRA · iteration 5 · Top

Sources: W:/Temp/opencode/glimmer_meshes_20261102/tuned_iter{1,5}/renders/{front34,side,top}.png